INCH-POUND

AN844 Rev 14
12 March 2020
SUPERSEDING
AN844 Rev 13
3 October 2013

DETAIL SPECIFICATION SHEET

ELBOW - PIPE TO HOSE, 45 DEGREE

Inactive for new design after 7 June 2011. For new design, use SAE-AS5186.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4843/2.

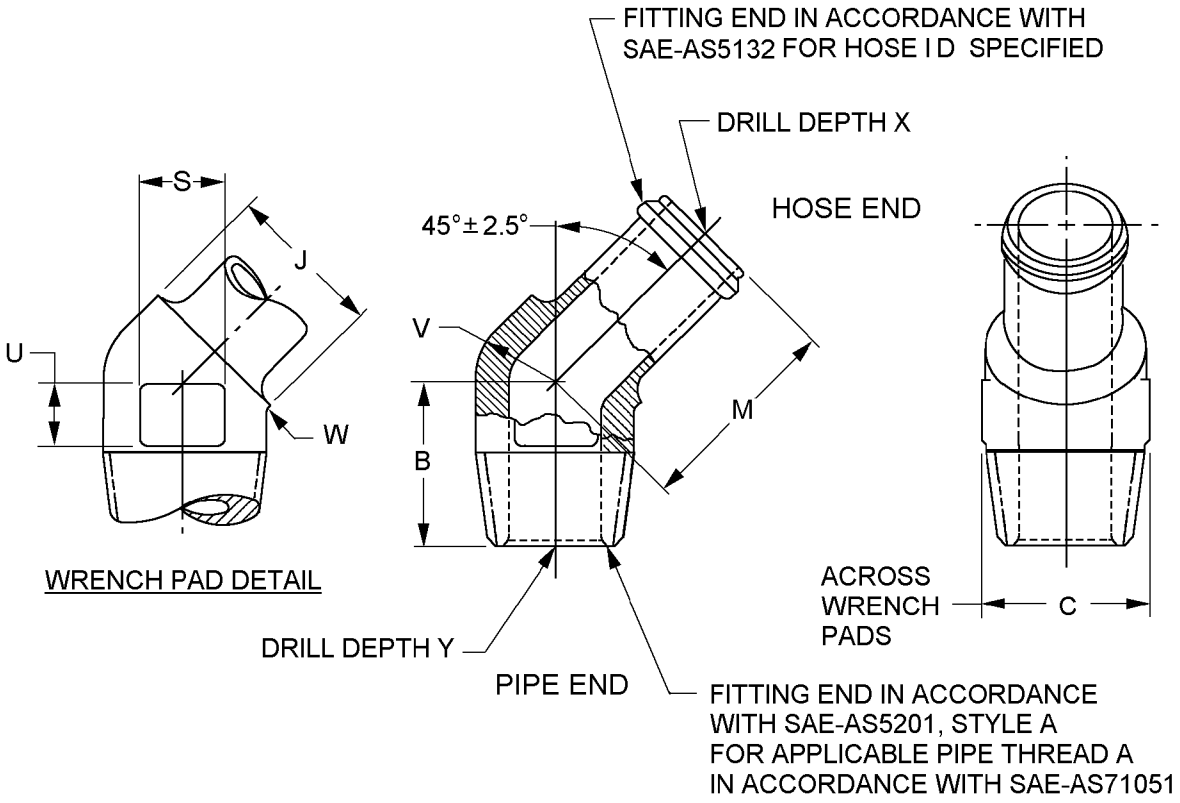


FIGURE 1. 45° elbow dimensions and configuration.

AMSC N/A

FSC 4730

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Dash number	Hose ID inches (mm)	A Pipe thread SAE-AS71051	B +.047 (1.19) -.000 inches (mm)	C inches (mm)
4	.250 (6.35)	1/8-27 ANPT	.625 (15.88)	.438 (11.13)
6	.375 (9.53)	1/4-18 ANPT	.875 (22.23)	.625 (15.88)
8	.500 (12.70)	3/8-18 ANPT	.922 (23.42)	.750 (19.05)
10	.625 (15.88)	1/2-14 ANPT	1.156 (29.36)	.938 (23.83)
12	.750 (19.05)	3/4-14 ANPT	1.203 (30.56)	1.125 (28.58)
16	1.000 (25.40)	3/4-14 ANPT	1.203 (30.56)	1.125 (28.58)
17	1.000 (25.40)	1-11.5 ANPT	1.484 (37.69)	1.375 (34.93)
20	1.250 (31.75)	1/ 1/4-11.5 ANPT	1.500 (38.10)	1.750 (44.45)
21	1.250 (31.75)	1 -11.5 ANPT	1.484 (37.69)	1.375 (34.93)
24	1.500 (38.10)	1 1/2-11.5 ANPT	1.563 (39.70)	2.000 (50.80)
25	1.500 (38.10)	1 1/4-11.5 ANPT	1.438 (37.67)	1.750 (44.45)

Dash number	J dia. inches (mm)	M +.047 (1.19) -.000 inches (mm)	S approx inches (mm)	U approx inches (mm)
4	.438 (11.13)	1.625 (41.28)	.313 (7.95)	.219 (5.56)
6	.578 (14.68)	1.688 (42.88)	.375 (9.53)	.281 (7.14)
8	.719 (18.26)	1.719 (43.66)	.500 (12.70)	.313 (7.95)
10	.891 (22.63)	1.781 (45.24)	.625 (15.88)	.375 (9.53)
12	1.094 (48.36)	1.813 (46.05)	.625 (15.88)	.438 (11.13)
16	1.094 (48.36)	1.813 (46.05)	.625 (15.88)	.438 (11.13)
17	1.375 (34.93)	1.875 (47.63)	.750 (19.05)	.563 (14.30)
20	1.719 (43.66)	1.938 (49.23)	.938 (23.83)	.563 (14.30)
21	1.375 (34.93)	1.875 (47.63)	.750 (19.05)	.563 (14.30)
24	1.953 (49.61)	2.000 (50.80)	1.125 (28.58)	.625 (15.88)
25	1.719 (43.66)	2.000 (50.80)	1.125 (28.58)	.625 (15.88)

FIGURE 1. 45° elbow dimensions and configuration - Continued.

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Dash number	V radius. inches (mm)	W radius inches (mm)	X +.047 (1.19) -.000 inches (mm)	Y +.047 (1.19) -.000 inches (mm)
4	.219 (5.56)	.063 (1.60)	1.672 (42.47)	.656 (16.66)
6	.282 (7.16)	.063 (1.60)	1.750 (44.45)	.938 (23.83)
8	.359 (9.12)	.094 (2.39)	1.797 (45.64)	1.000 (25.40)
10	.438 (11.13)	.094 (2.39)	1.890 (48.10)	1.266 (32.16)
12	.547 (13.89)	.125 (3.18)	1.938 (49.23)	1.359 (34.52)
16	.547 (13.89)	.125 (3.18)	2.000 (50.80)	1.359 (34.52)
17	.688 (17.48)	.125 (3.18)	2.063 (52.40)	1.672 (42.47)
20	.859 (21.82)	.125 (3.18)	2.156 (54.76)	1.719 (43.66)
21	.688 (17.48)	.125 (3.18)	2.109 (53.57)	1.672 (42.47)
24	.969 (24.61)	.125 (3.18)	2.281 (57.94)	1.844 (46.84)
25	.859 (21.82)	.125 (3.18)	2.281 (57.94)	1.656 (42.06)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.016 inch (0.41 mm).
4. Break sharp edges .and remove all hanging burrs and slivers.
5. Machined surfaces shall be finished to 125µ in Ra, forged surfaces shall be 250µ inches Ra, unless otherwise specified on the figures. Surface finish shall be in accordance with ASME B46.1.
6. For design features purposes, this standard takes precedence over documents referenced herein.
7. Referenced documents shall be of the issue in effect on date of invitation for bid.

FIGURE 1. 45° elbow dimensions and configuration - Continued.

REQUIREMENTS:

The elbow dimensions and configuration shall be in accordance with figure 1.

Materials and finishes shall be in accordance with SAE-AS4843/2, see table I for material and finish code.

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TABLE I. Material and finish codes.

Material and finish code	Material	Protective chemical finish 3/ 4/
No code 1/	Copper alloy, type 377 forging in accordance with ASTM B124/B124M or half hard forging or bar in accordance with ASTM B138/B138M or bar in accordance with SAE-AMS4614.	No finish.
BC 1/	Copper alloy, type 377 forging in accordance with ASTM B124/B124M or half hard forging or bar in accordance with ASTM B138/B138M or bar in accordance with SAE-AMS4614.	Cadmium in accordance with SAE-AMS-QQ-P-416, type II, class 3.
D 2/	Type 2014-T6 aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 or SAE-AMS4133, or type 2024-T6 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6, or type 2024-T851 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6 or SAE-AMS4339.	Anodize in accordance with SAE-AMS2472 or MIL-A-8625, type II, class 2, dye blue, duplex seal in accordance with procurement specification.
DV 2/	Type 2014-T6 aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 or SAE-AMS4133, or type 2024-T6 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6, or type 2024-T851 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6 or SAE-AMS4339.	High purity aluminum in accordance with MIL-DTL-83488, class 3, type II with maximum coating thickness of .0005 inch. Glass bead peen pressure shall be 25 psi (1.72 bar) maximum.
J	Type 304 corrosion resistant steel forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5639.	Passivate in accordance with SAE-AMS2700, type VI or VII.
K	Type 316 corrosion resistant steel forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5648.	Passivate in accordance with SAE-AMS2700, type VI or VII.
R	Type 321 corrosion resistant steel forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5645.	Passivate in accordance with SAE-AMS2700, type VI or VII.

See notes at end of table

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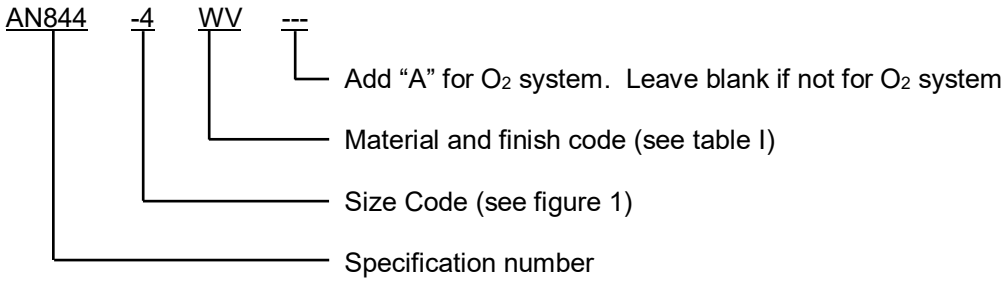
TABLE I. Material and finish codes - Continued.

Material and finish code	Material	Protective chemical finish 3/ 4/
W 2/	Type 7075-T73 aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 or SAE-AMS4141, or type 7075-T73 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/9, in accordance with type 7075-T7351 Aluminum alloy bar in accordance with SAE-AMS4124.	Anodize in accordance with SAE-AMS2472 or MIL-A-8625, type II, class 2, dye brown similar to color SAE-AMS-STD-595/10080, duplex seal in accordance with procurement specification.
WV 2/	Type 7075-T73 aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 or SAE-AMS4141, or type 7075-T73 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/9, in accordance with type 7075-T7351 Aluminum alloy bar in accordance with SAE-AMS4124.	High purity aluminum in accordance with MIL-DTL-83488, class 3, type II with maximum coating thickness of .0005 inch. Glass bead peen pressure shall be 25 psi (1.72 bar) maximum.
T 3/	Titanium	Anodize in accordance with SAE-AMS2488, type 2.

1/ Material code was dash on previous revisions, changed to agree with SAE-ARP1590.
2/ Aluminum code D is inactivated, use code W.
3/ Titanium and cadmium shall not be used in oxygen systems.
4/ Cadmium shall not be used in potable water systems.

Oxygen systems for aerospace, shipboard, and ground support equipment. Parts for use on oxygen systems shall be identified in the PIN as code “A” and shall be furnished cleaned, packaged, and labeled in accordance with SAE-AS611 to a process approved by the user.

Part or Identifying Number (PIN): The PIN consists of the prefix “AN”, specification sheet number, dash number for tube and port size, material finish code letter(s) and blank of “A” for oxygen cleaning requirements. Unassigned PIN’s shall not be used.



PIN examples:

AN844-4W indicates a 45° elbow 1/4 inch tube and hose size, aluminum alloy 7075-T73.

AN844-4WV indicates a 45° elbow 1/4 inch tube and hose size, aluminum alloy 7075-T73 finish with high purity aluminum.

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AN844-4WA indicates a 45° elbow 1/2 inch tube and hose size, aluminum alloy 7075-T73 for use on oxygen systems.

Guidance on use of alternative parts with less hazardous or non-hazardous materials. This specification provides for a number of alternative plating materials via the PIN. Users should select the PIN with the least hazardous material that meets the form, fit, and function requirements of their application.

Supersession data. The aluminum “D” designator has been replaced by the “W” designator.

Marking: Part shall be permanently marked with the AN PIN, and include the manufacturers CAGE, name, or trademark.

Table II provides a detailed cross-reference of AN844 PINs and replacement SAE-AS5186 PINs. Users are cautioned to evaluate replacements for their particular application.

CAUTION: The superseding information is valid as of the date of this specification and may be superseded by subsequent revisions of the superseding document.

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TABLE II. Cross-reference data. 1/

AN PIN Inactive part	Hose Size	Pipe Size	AS PIN 2/ Active part
AN844-4	0.250	0.125	AS5186B0402
AN844-4D	0.250	0.125	AS5186W0402
AN844-4R	0.250	0.125	AS5186R0402
AN844-4S	0.250	0.125	AS5186R0402
AN844-4W	.2500	0.125	AS5186W0402
AN844-6	0.375	0.250	AS5186B0604
AN844-6D	0.375	0.250	AS5186W0604
AN844-6R	0.375	0.250	AS5186R0604
AN844-6S	0.375	0.250	AS5186R0604
AN844-6W	0.375	0.250	AS5186W0604
AN844-8	0.500	0.375	AS5186B0806
AN844-8D	0.500	0.375	AS5186W0806
AN844-8R	0.500	0.375	AS5186R0806
AN844-8S	0.500	0.375	AS5186R0806
AN844-8W	0.500	0.375	AS5186W0806
AN844-10	0.625	0.500	AS5186B1008
AN844-10D	0.625	0.500	AS5186W1008
AN844-10R	0.625	0.500	AS5186R1008
AN844-10S	0.625	0.500	AS5186R1008
AN844-10W	0.625	0.500	AS5186W1008
AN844-12	0.750	0.750	AS5186B1212
AN844-12D	0.750	0.750	AS5186W1212
AN844-12R	0.750	0.750	AS5186R1212
AN844-12S	0.750	0.750	AS5186R1212
AN844-12W	0.750	0.750	AS5186W1212
AN844-16	1.000	0.750	AS5186B1612
AN844-16D	1.000	0.750	AS5186W1612
AN844-16R	1.000	0.750	AS5186R1612
AN844-16S	1.000	0.750	AS5186R1612
AN844-16W	1.000	0.750	AS5186W1612
AN844-17	1.000	1.000	AS5186B1616
AN844-17D	1.000	1.000	AS5186W1616
AN844-17R	1.000	1.000	AS5186R1616
AN844-17S	1.000	1.000	AS5186R1616
AN844-17W	1.000	1.000	AS5186W1616
AN844-20	1.250	1.250	AS5186B2020
AN844-20D	1.250	1.250	AS5186W2020
AN844-20R	1.250	1.250	AS5186R2020
AN844-20S	1.250	1.250	AS5186R2020
AN844-20W	1.250	1.250	AS5186W2020

See notes at end of table.

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TABLE II. Cross-reference data - Continued. 1/

AN PIN Inactive part	Hose Size	Pipe Size	AS PIN 2/ Active part
AN844-21	1.250	1.000	AS5186B2016
AN844-21D	1.250	1.000	AS5186W2016
AN844-21R	1.250	1.000	AS5186R2016
AN844-21S	1.250	1.000	AS5186R2016
AN844-21W	1.250	1.000	AS5196W2016
AN844-24	1.500	1.500	AS5186B2424
AN844-24D	1.500	1.500	AS5186W2424
AN844-24R	1.500	1.500	AS5186R2424
AN844-24S	1.500	1.500	AS5186R2424
AN844-24W	1.500	1.500	AS5186W2424
AN844-25	1.500	1.250	AS5186B2420
AN844-25D	1.500	1.250	AS5186W2420
AN844-25R	1.500	1.250	AS5186R2420
AN844-25S	1.500	1.250	AS5186R2420
AN844-25W	1.500	1.250	AS5186W2420

1/ For new design use material designator R and W.

2/ SAE “B” designator for copper alloy parts can be either “no finish” or “cadmium plated”. This makes determination of hazardous materials more difficult and disposing of hazardous materials more laborious.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

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Referenced documents. In addition to SAE-AS4843/2, this document references the following:

MIL-A-8625	SAE-AMS-STD-595/10080	SAE-AMS5645
MIL-DTL-83488	SAE-AMS2472	SAE-AMS5648
ASME B46.1	SAE-AMS2488	SAE-ARP-1590
ASTM B124/B124M	SAE-AMS2700	SAE-AS611
ASTM B138/B138M	SAE-AMS4124	SAE-AS5132
SAE-AMS-QQ-A-225/6	SAE-AMS4133	SAE-AS5186
SAE-AMS-QQ-A-225/9	SAE-AMS4141	SAE-AS5201
SAE-AMS-QQ-A-367	SAE-AMS4339	SAE-AS71051
SAE-AMS-QQ-P-416	SAE-AMS4614	
SAE-AMS-QQ-S-763	SAE-AMS5639	

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2020-037)

Review activities:
Army - MI

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